

Starter-Motor Risk Prediction Management Brief

The one-read summary: what the system catches, the alert lead time,
the maintenance-window verdict, what we honestly miss, and the asks.

0.9321

Ranking AUROC

13 / 14

Failures caught

0 / 20

Battery-alert false alarms

Deploy

On existing telematics

MODEL

V1.1_SM (frozen)

FLEET

34 trucks (14 F + 20 NF)

AUDIENCE

DICV management

DATE

02 July 2026

COMPANION

Full validation report (same folder)

STATUS

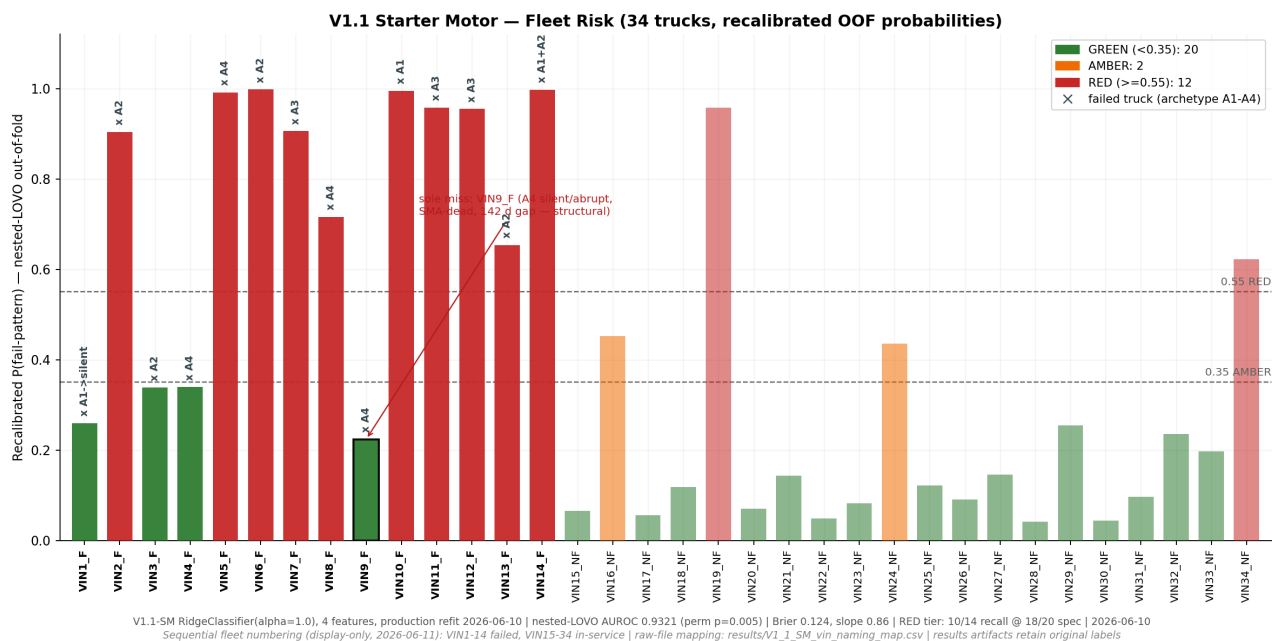
Final

The system in one read

0.9321 Ranking AUROC (LOVO)	13 / 14 Failures caught	0 / 20 Battery-alert false alarms	168 d Median alert lead
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The starter-motor risk-prediction system **ranks** every truck by failure risk from telemetry you already collect (crank + resting voltage), **alerts** when a degradation precursor persists, and **schedules** service inside a validated maintenance window — turning surprise breakdowns into planned maintenance. No new hardware.

The four numbers that matter	Value	Meaning
Ranking AUROC (LOVO)	0.9321	Orders failed above healthy — 261 of 280 pairs, unseen trucks
Failures caught	13 / 14	Alert channels flag all but the one structural blind spot
Median alert lead	168 d	First-fire lead before recorded failure (range 77–424)
Battery-alert false alarms	0 / 20	A2 battery-cascade detector never flags a healthy truck



Fleet risk picture — all 34 trucks; failed trucks separate cleanly from the healthy band (frozen V1.1_SM, nested LOVO). Non-failed trucks shown under the graph's native sequential display numbering (VIN15_NF–VIN34_NF).

What to say in the room

“93% ranking accuracy — 13 of 14 failures caught, median 168 days' warning, with zero battery-alert false alarms, using data we already have.” Say *ranking* accuracy, not bare accuracy.

Detection, lead time and the window verdict

What the 0.9321 means. Given a random failed and a random healthy truck, the model ranks the failed one higher 93.2% of the time — 261 of 280 truck-pairs, measured on trucks it never saw in training (selection optimism just one pair: 262 vs 261). It is *ranking*, not “93% of trucks correct.”

Per-truck alert lead (earliest validated channel, vs recorded failure date):

Truck	Tier	Earliest lead (d)	Truck	Tier	Earliest lead (d)
VIN5_F_SM	RED	424	VIN3_F_SM	GREEN	168
VIN13_F_SM	RED	301	VIN6_F_SM	RED	168
VIN11_F_SM	RED	266	VIN10_F_SM	RED	160
VIN7_F_SM	RED	266	VIN8_F_SM	RED	135
VIN14_F_SM	RED	245	VIN12_F_SM	RED	128
VIN1_F_SM	GREEN	232	VIN4_F_SM	GREEN	125
			VIN2_F_SM	RED	77
			VIN9_F_SM	GREEN	MISSED

Maintenance-window verdict

Day-precise RUL is false precision here (survival error 576 d vs 44 d for a constant), so we ship a **tier + window**. Of the 11 windowed failures, **9 fell inside** the predicted window and both misses were late on the safe side. Windows: battery 28–91 days, persistence 126–284 days.

Limits, and what we ask for

The honest limits. The binding constraint is 14 failure events: the AUROC confidence interval is wide ([0.811, 0.986]) and a handful of trucks decide every threshold. One truck (VIN9-class) is a structural blind spot — telemetry-dead configuration, a 142-day silent gap and an abrupt mode with no precursor; five of the fourteen failures carry 32–142 days of telemetry silence before the recorded failure date; and failure dates come from workshop records, so lead times carry timing uncertainty. Scores correlate with observation length, but this is label-mediated and defended by a fixed-window control (zero AUROC borrowed) and a decay-to-chance horizon curve. Every one of these is a **data ceiling, not a method ceiling**.

- **Deploy now** on existing telematics: weekly risk ranking + green/amber/red tiers, the persistence and A2 battery-cascade alerts, and tier-based maintenance windows.
- **Instrument for the next step:** cranking current / battery SoC-SoH and ≥ 1 Hz crank-voltage logging (the biggest unlocks), plus maintenance & parts records to create supervised labels.
- **Scale to 500 trucks (Phase 2)** for ~30–50+ failure events — the unlock for per-truck timing and sharper tiers. Same validated method, no new risk.

Bottom line

Validated, honest and deployable today — planned starter/battery maintenance from data you already collect, with no new hardware. The 0.9321 ranking is real and conservative, the 13-of-14 detection is actionable, and it improves with scale.